

Ocean science

The science of oceanography began with the observations of naturalists like Charles Darwin, and continued with the scientific voyage of HMS *Challenger* in the late 19th century. This work laid the foundation for modern research, using ships, submersibles, and even satellites to relay data to oceanographic study centers. Modern oceanography covers all aspects of the oceans, from ocean-floor geology to the causes of oceanic storms.



National Oceanography Center in Southampton

OCEAN SCIENCE

Oceanography is one of the most complex sciences, involving physics, chemistry, geology, marine biology, and meteorology. These subjects are studied at oceanographic research institutes associated with universities, such as those at Southampton in England, Naples in Italy, and the Woods Hole Oceanographic Institution in the US. These institutes operate research ships of their own, including this one docked at Southampton.

RESEARCH SHIPS

Oceanographic research ships are specially built for the job. As well as having laboratories, sampling equipment, and surveying gear, many carry deep-sea submersibles.

These need special handling equipment, shown in action here as the Woods Hole research ship *Atlantis* hoists the submersible *Alvin* from the water after a dive.



OCEAN-FLOOR DRILLING

The nature of the ocean floor has been probed by deep-sea drilling, which collects samples of the rocks to build up a picture of its geology. The Japanese drilling ship *Chikyu*, shown here, can drill to the amazing depth of 23,000 ft (7,000 m) below the ocean floor, in water 8,200 ft (2,500 m) deep. The data collected by these drilling projects has changed our understanding of the planet.

